THE LIMIT OF A FUNCTION

INTRODUCTION TO CALCULUS

prepared by:

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Electronics Engineer

GRADING SYSTEM

FINAL GRADE

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ENDTERM

TERM GRADE

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EXAM

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SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

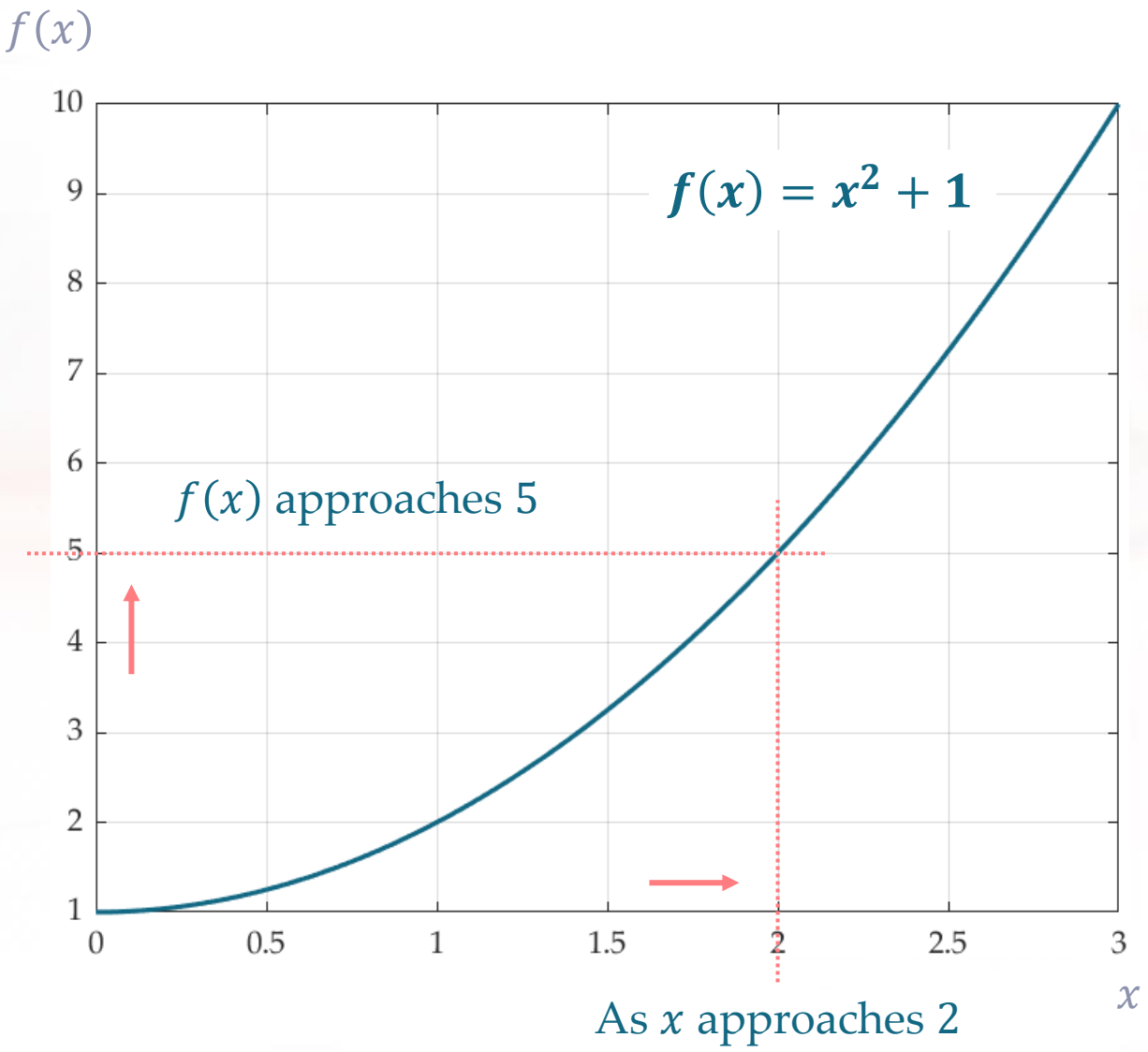
The Limit of a Function

Limit Laws

- Sum, Difference, Constant Multiple
- Product, Quotient, Power, Root

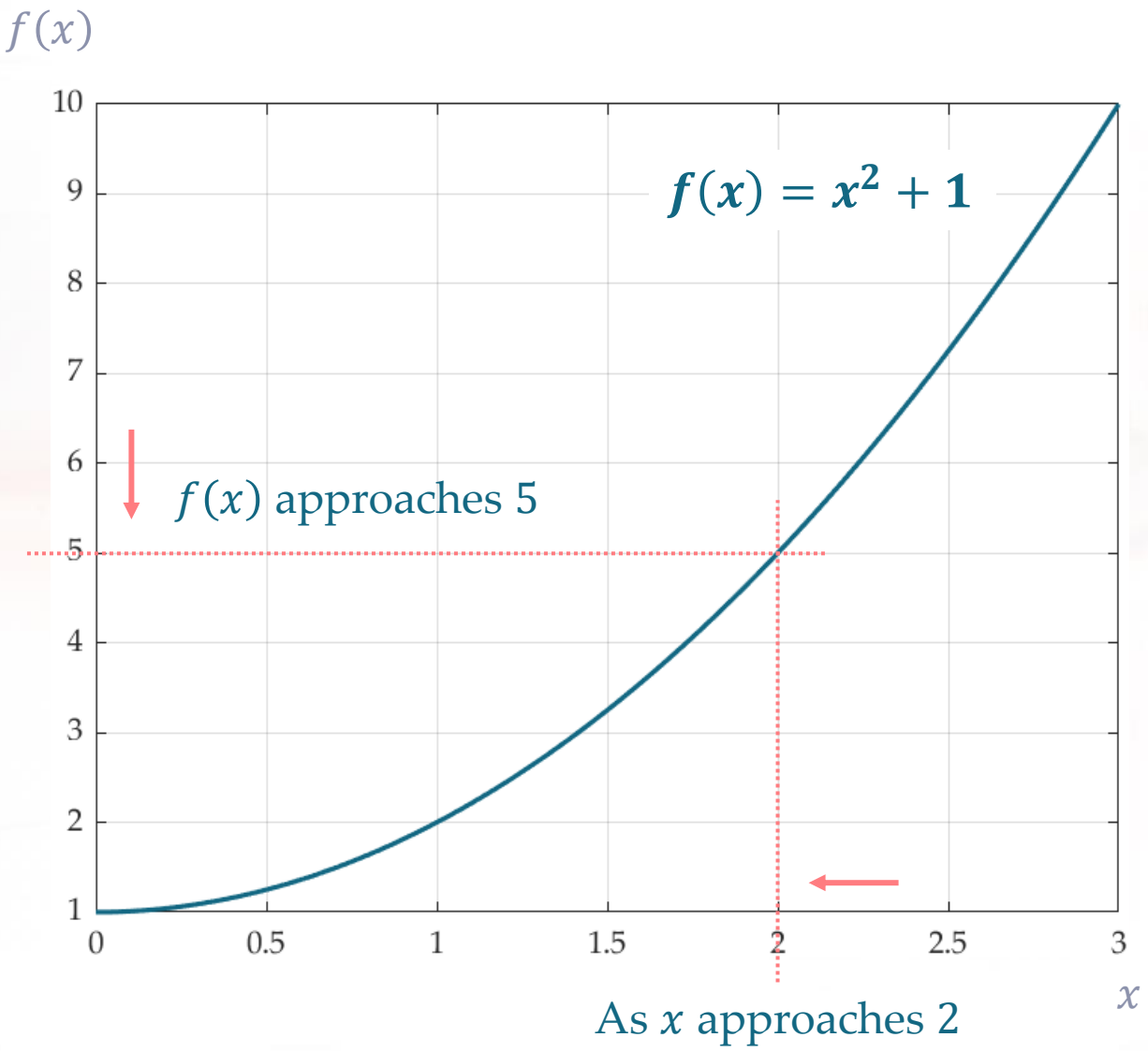
THE LIMIT OF A FUNCTION

PARABOLA



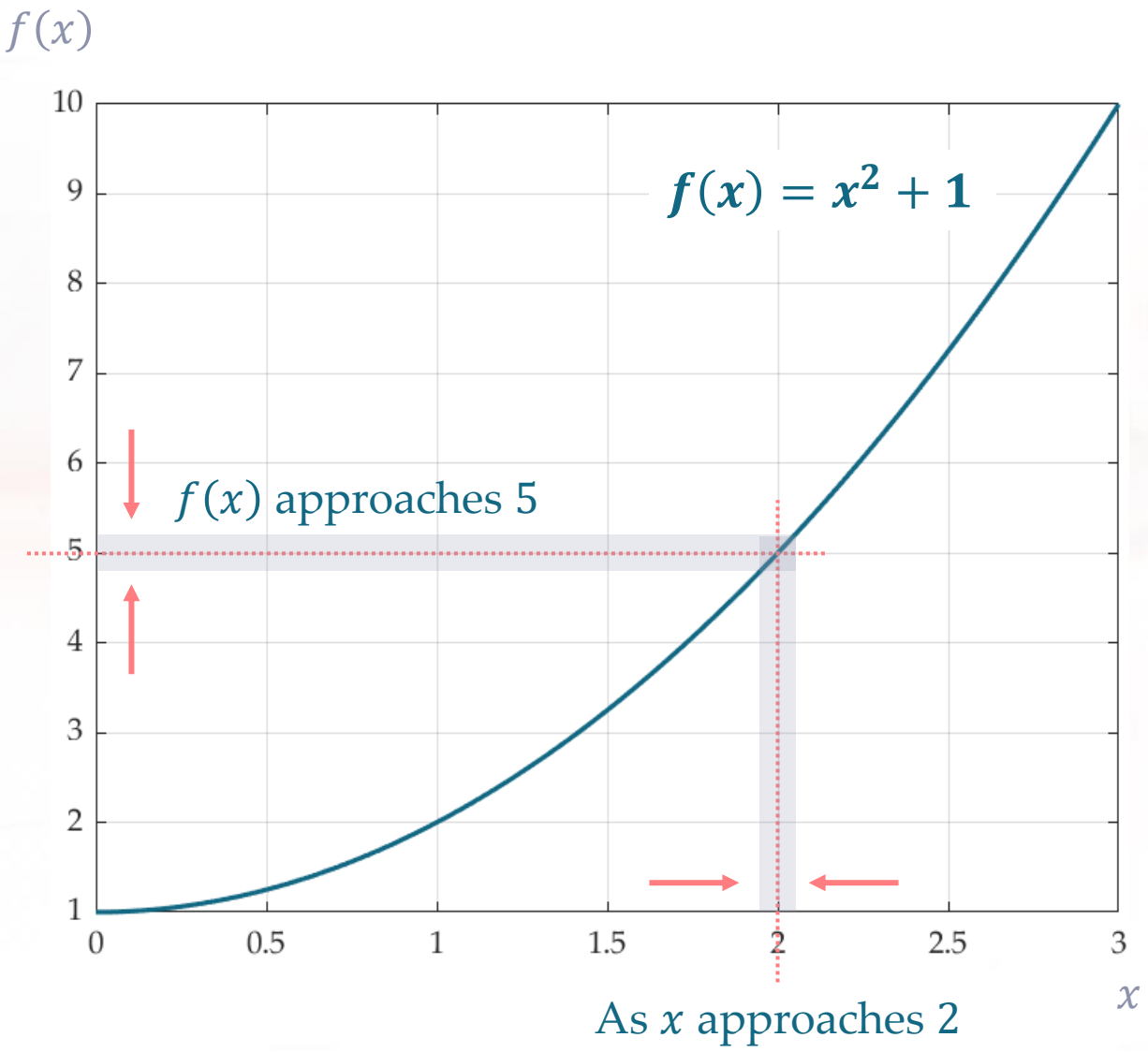
x	$f(x)$
0	1
1	2
1.5	3.25
1.8	4.24
1.9	4.61
1.91	4.6481
1.92	4.6864
1.93	4.7249
1.95	4.8025
1.96	4.8416
1.97	4.8809
1.98	4.9204
1.99	4.9601
2	5

PARABOLA



x	$f(x)$
3	10
2.5	7.25
2.3	6.29
2.1	5.41
2.09	5.3681
2.08	5.3264
2.07	5.2849
2.06	5.2436
2.05	5.2052
2.04	5.1616
2.03	5.1209
2.02	5.0804
2.01	5.0401
2	5

PARABOLA



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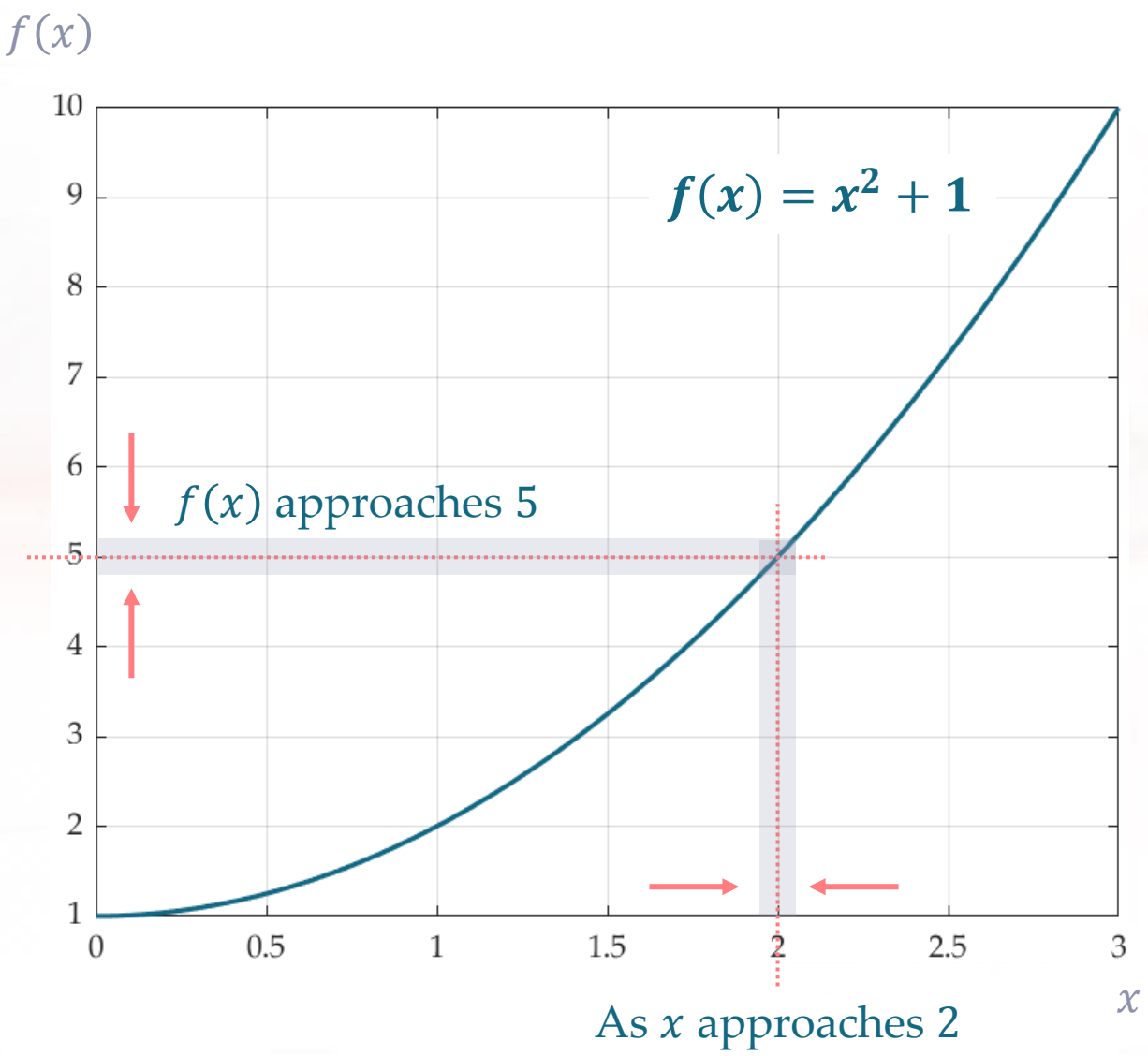
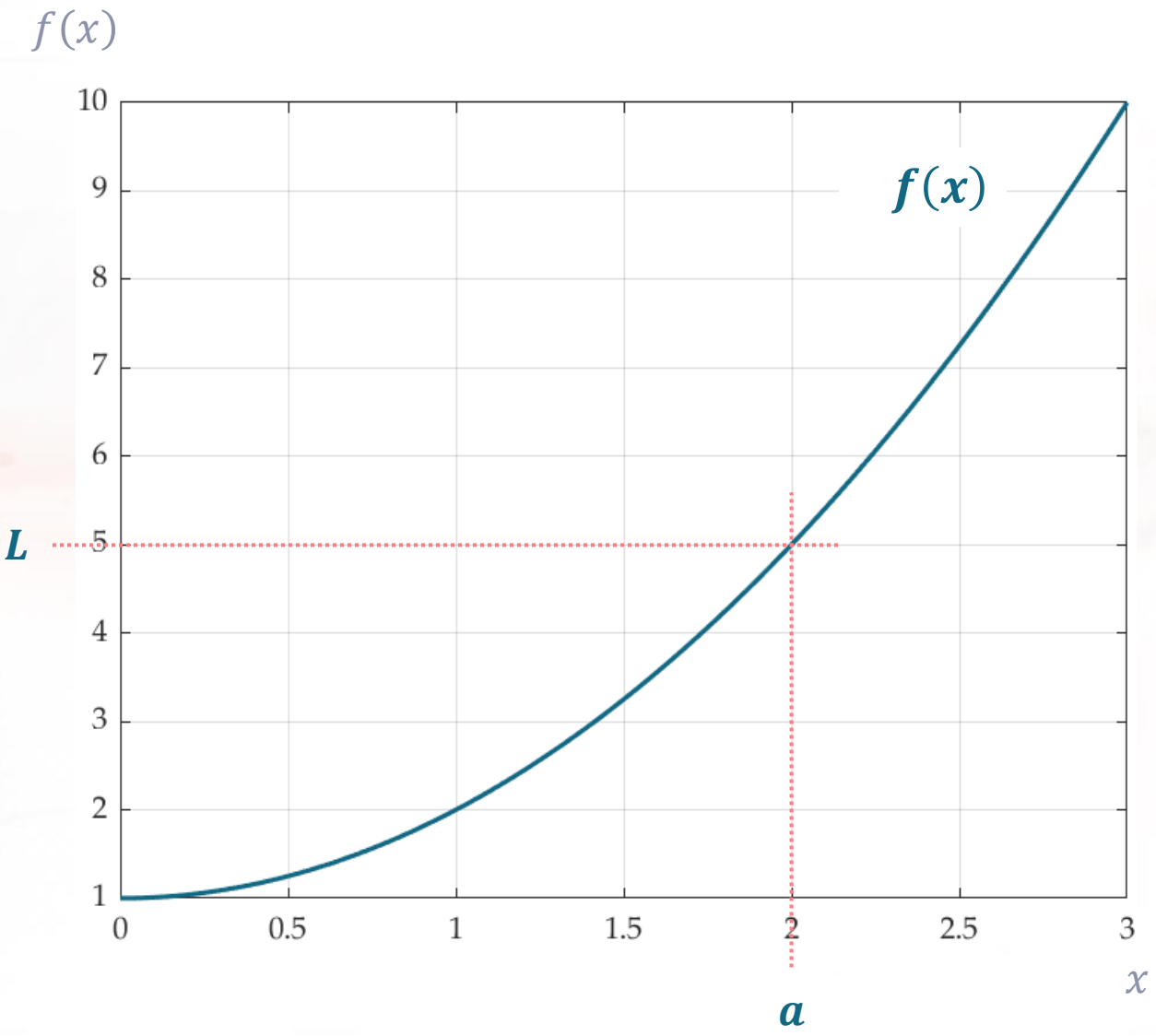
2	5



Notation

$$\lim_{x \to 2} \left(x^2 + 1\right) = 5$$

DEFINITION OF LIMIT

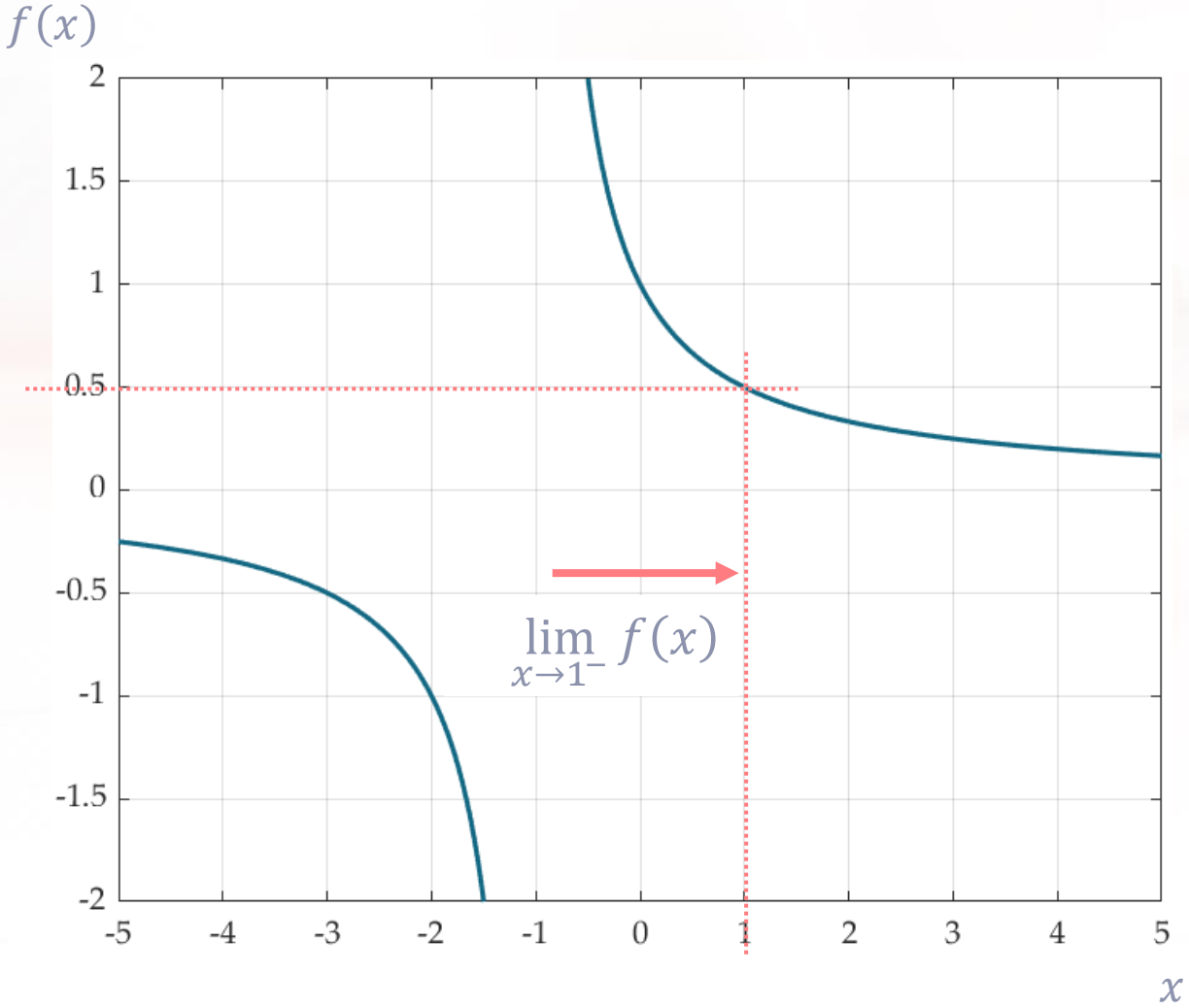


Suppose $f(x)$ is defined when x is **near** the number a, then

$$\lim_{x \to a} f(x) = L$$

"the limit of $f(x)$, as x approaches a, equals L"

Determine $\lim_{x \to 1} \frac{x-1}{x^2-1}$

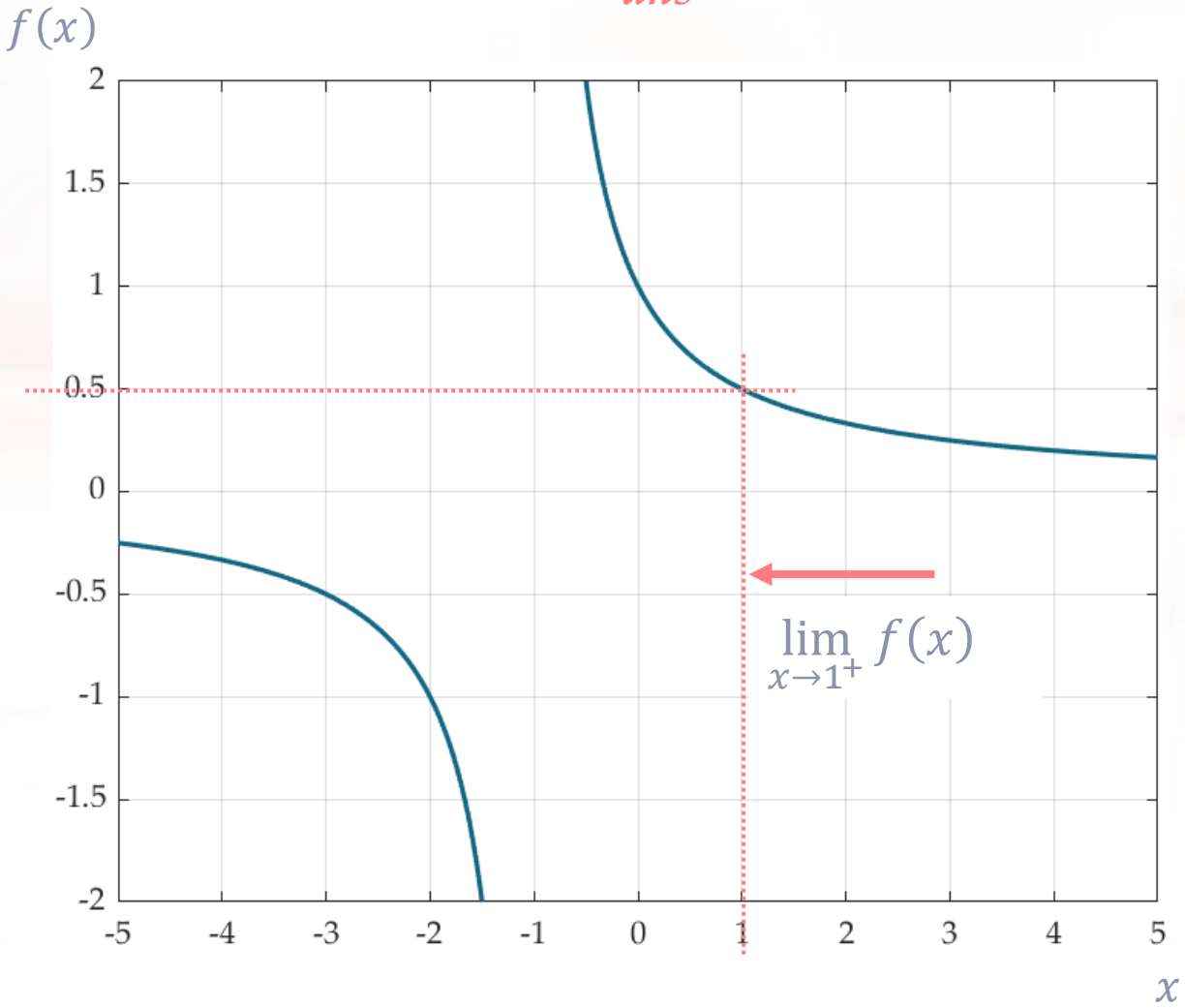


As x approaches 1 from the **left**

x	$f(x)$
0	1
0.5	0.66667
0.6	0.625
0.7	0.58824
0.8	0.55556
0.9	0.52632
0.95	0.51282
0.96	0.5102
0.97	0.50761
0.98	0.50505
0.99	0.50251
1	∞

Determine $\lim_{x \to 1} \frac{x-1}{x^2-1}$ $\boxed{= 0.5}$

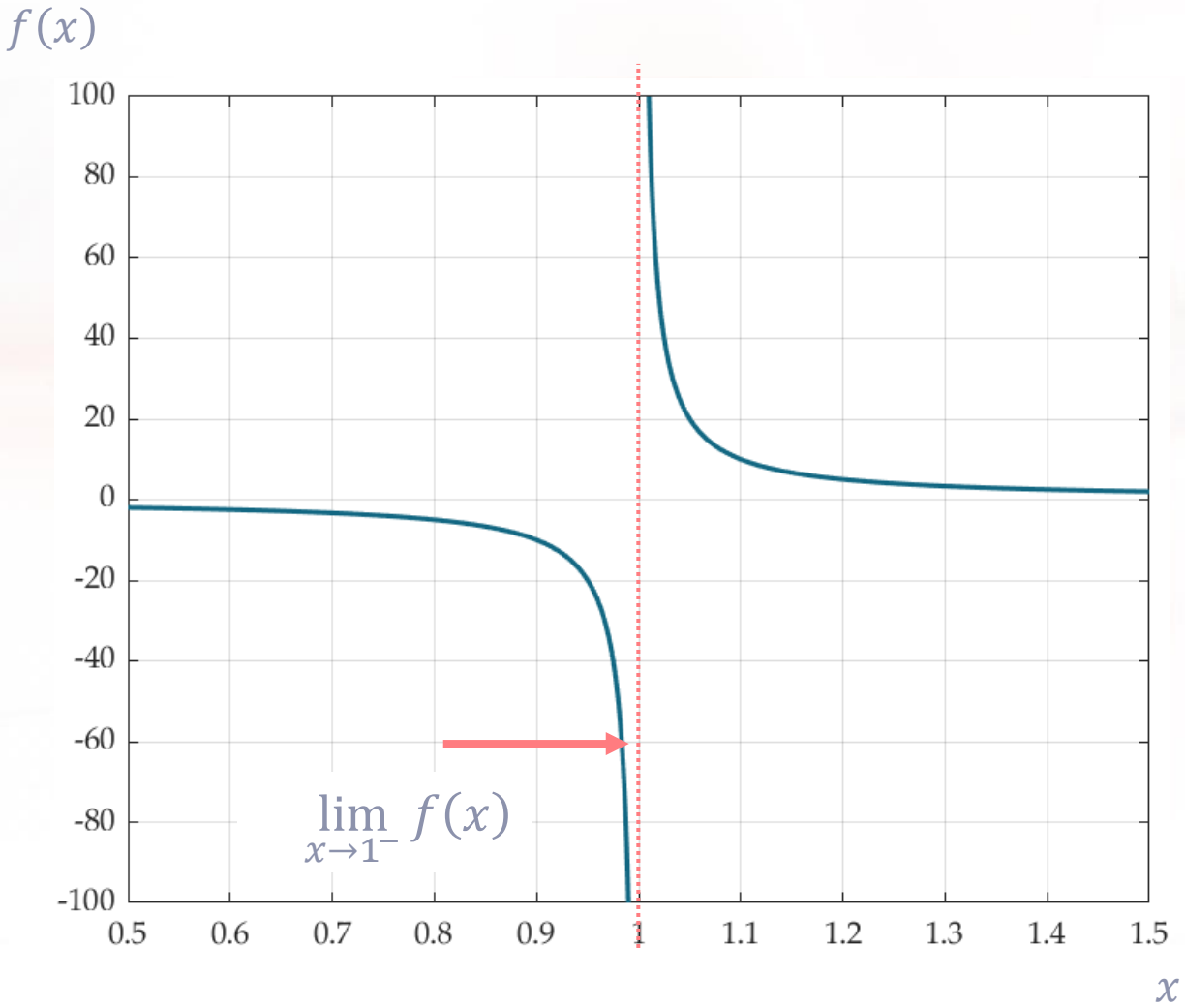
ans



As x approaches 1 from the **right**

x	$f(x)$
2.0	0.33333
1.5	0.4
1.4	0.41667
1.3	0.43478
1.2	0.45455
1.1	0.47619
1.05	0.4878
1.04	0.4902
1.03	0.49261
1.02	0.49505
1.01	0.49751
1	∞

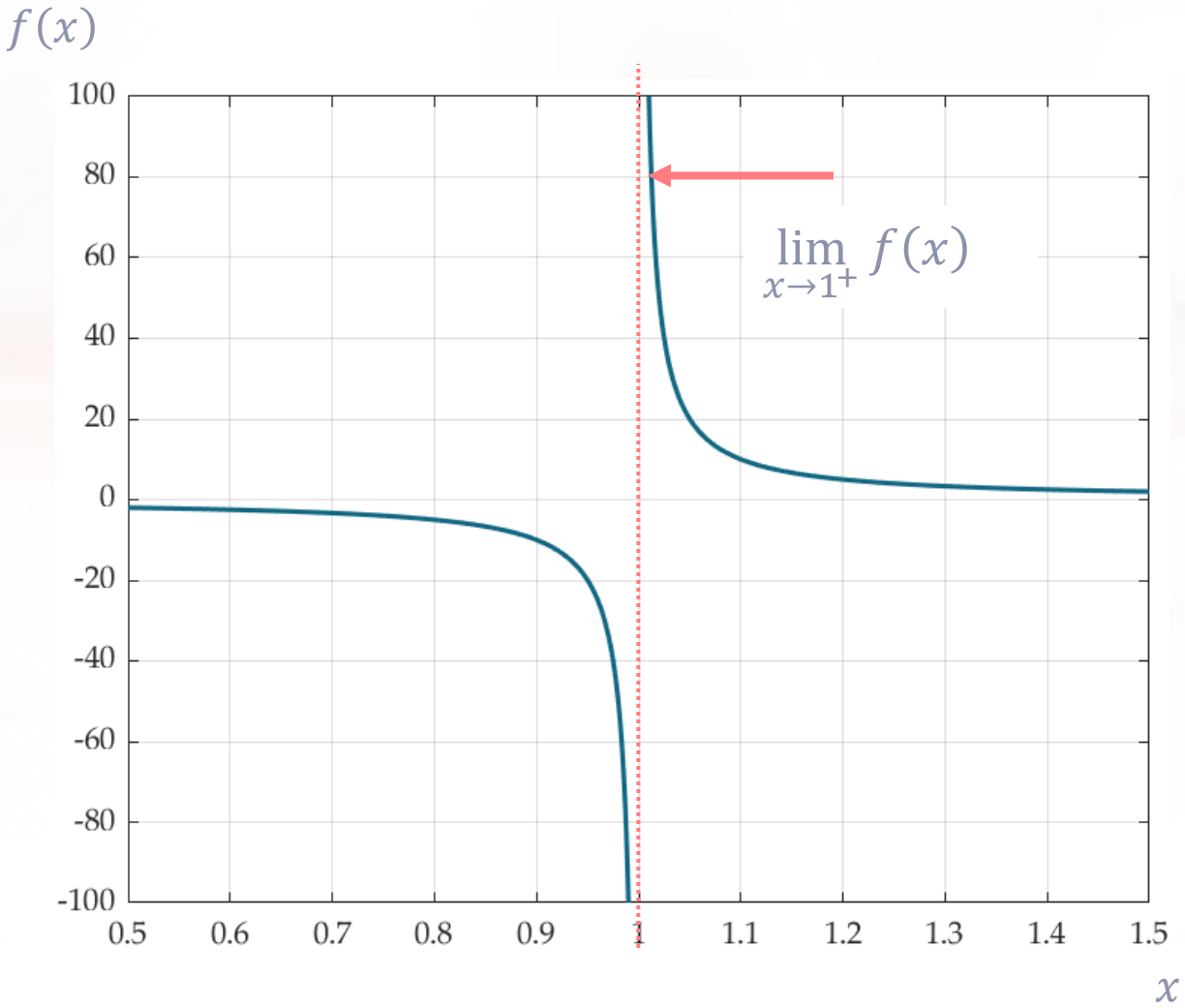
Determine $\lim_{x \to 1} \frac{1}{x-1}$



As x approaches 1 from the **left**

x	$f(x)$
0.9	-10
0.95	-20
0.99	-100
0.999	-1,000
0.9999	-10,000
0.99999	-100,000
0.999999	-1,000,000
$\to 1^-$	$-\infty$

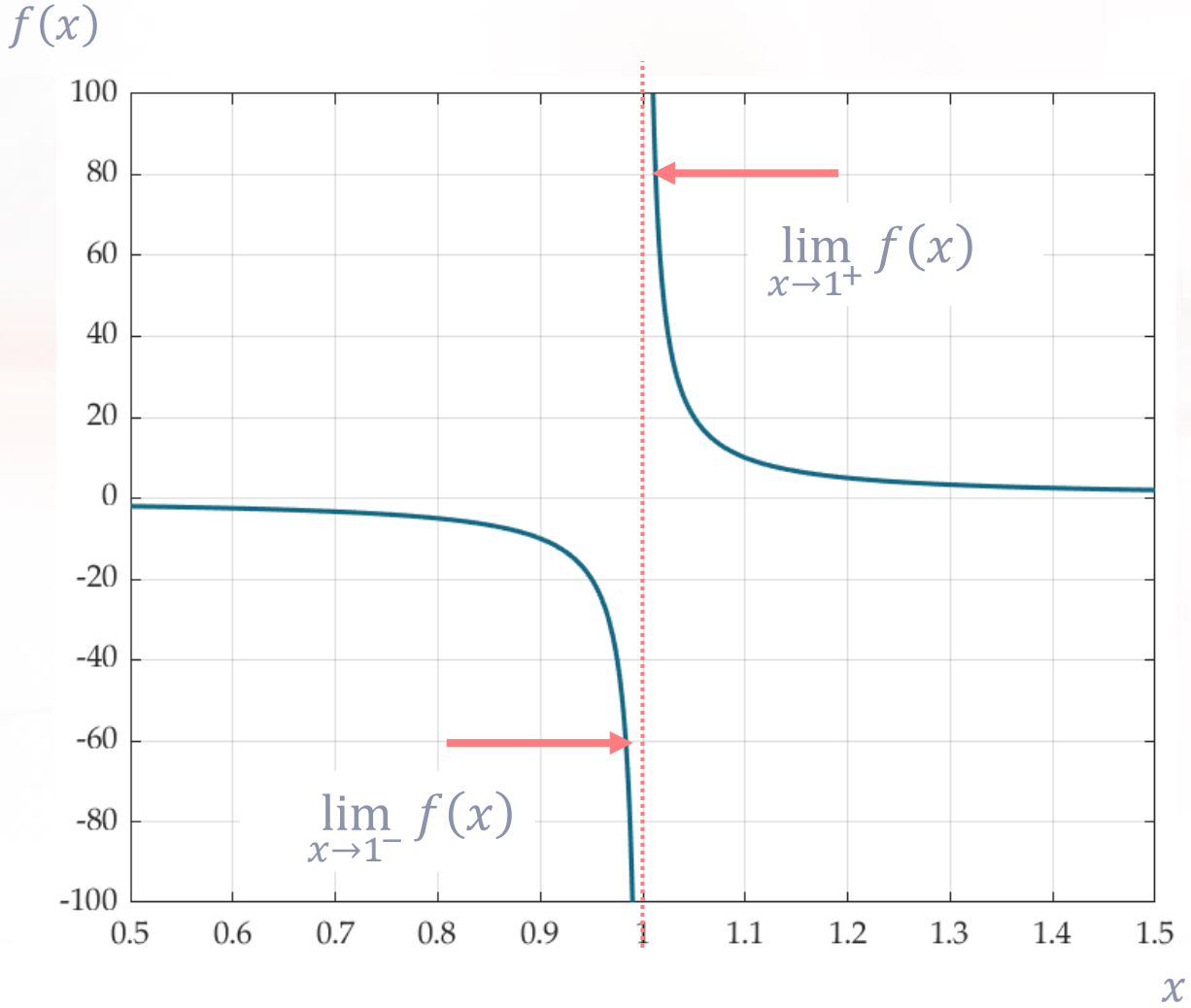
Determine $\lim_{x \to 1} \frac{1}{x-1}$



As x approaches 1 from the **right**

x	$f(x)$
1.1	10
1.05	20
1.01	100
1.001	1,000
1.0001	10,000
1.00001	100,000
1.000001	1,000,000
$\to 1^-$	$+\infty$

Determine $\lim_{x \to 1} \frac{1}{x-1}$



Left-hand limit

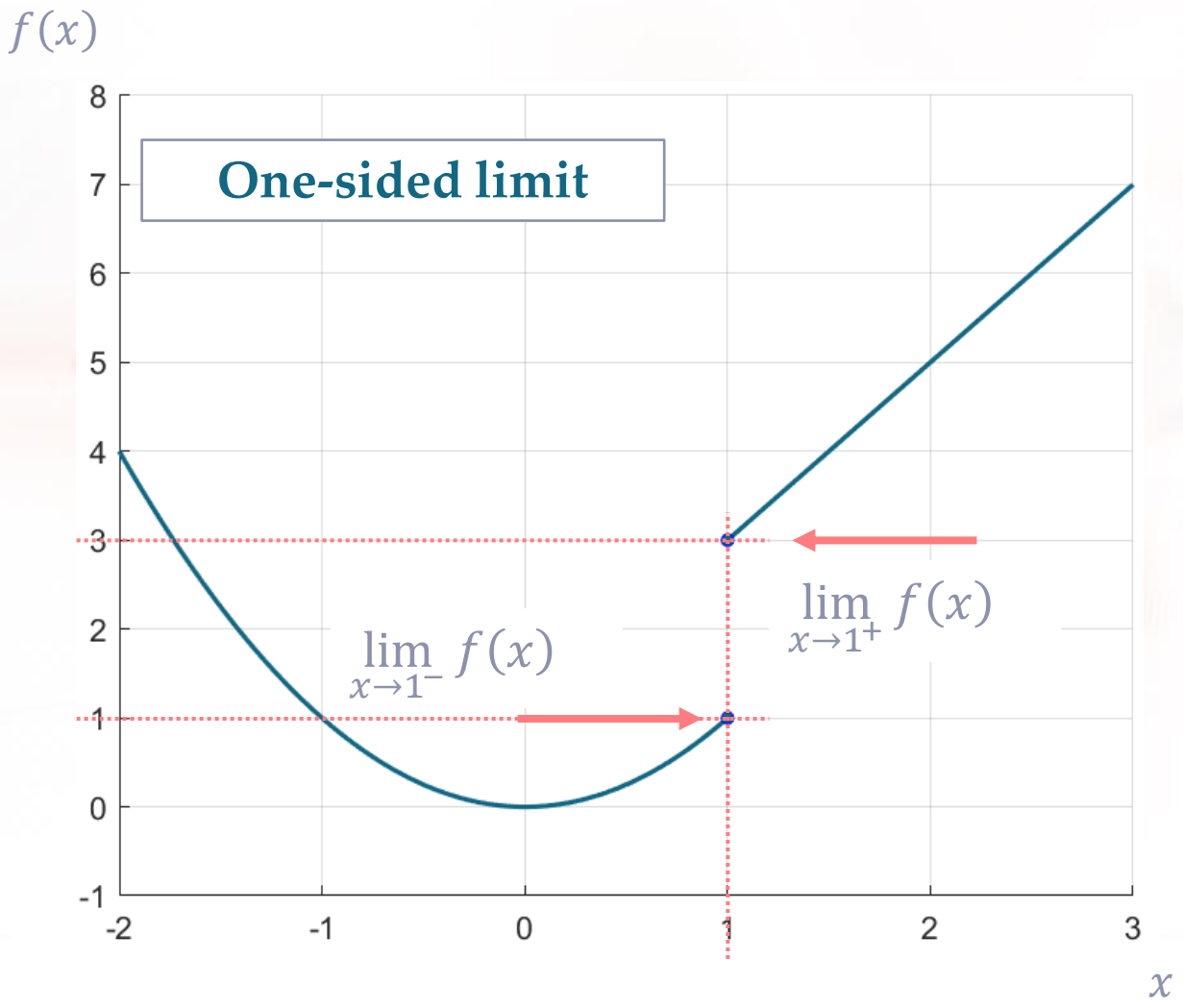
$$\lim_{x \to 1^-} f(x) = -\infty$$

Right-hand limit

$$\lim_{x \to 1^+} f(x) = \infty$$

The **limit does not exist**, since the left-hand limit and the right-hand limit are not equal

Determine $\lim_{x \to 1} \begin{cases} x^2, & \text{if } x < 1 \\ 2x + 1, & \text{if } x \geq 1 \end{cases}$



Left-hand limit

$$\lim_{x \to 1^-} f(x) = 1$$

Right-hand limit

$$\lim_{x \to 1^+} f(x) = 3$$

The **limit does not exist**, since the left-hand limit and the right-hand limit are not equal

LIMIT LAWS

$$\lim_{x \to a}[f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$$

Evaluate $\lim_{x \to 2}(x^2 + 1)$

$$\lim_{x \to 2}(x^2 + 1) = \lim_{x \to 2}(x^2) + \lim_{x \to 2}(1)$$

$$\lim_{x \to 2}(x^2 + 1) = 2^2 + 1$$

$$\lim_{x \to 2}(x^2 + 1) = 4 + 1$$

$$\boxed{\lim_{x \to 2}(x^2 + 1) = 5}$$

ans

DIFFERENCE LAW

$$\lim_{x \to a}[f(x) - g(x)] = \lim_{x \to a} f(x) - \lim_{x \to a} g(x)$$

Evaluate $\lim_{x \to 4}(x^2 - 3x)$

$$\lim_{x \to 4}(x^2 - 3x) = \lim_{x \to 4}(x^2) - \lim_{x \to 4}(3x)$$

$$\lim_{x \to 4}(x^2 - 3x) = 4^2 - 3(4)$$

$$\lim_{x \to 4}(x^2 - 3x) = 16 - 12$$

$$\boxed{\lim_{x \to 2}(x^2 - 3x) = 4}$$

ans

CONSTANT MULTIPLE LAW

$$\lim_{x \to a}[cf(x)] = c\lim_{x \to a} f(x)$$

Evaluate $\lim_{x \to 2}(5x^3)$

$$\lim_{x \to 2}(5x^2) = 5\lim_{x \to 2}(x^3)$$

$$\lim_{x \to 2}(5x^2) = 5(2^3)$$

$$\lim_{x \to 2}(5x^2) = 5(8)$$

$$\boxed{\lim_{x \to 2}(5x^3) = 40}$$

ans

PRODUCT LAW

$$\lim_{x\to a}[f(x)g(x)] = \lim_{x\to a} f(x) \cdot \lim_{x\to a} g(x)$$

Evaluate $\lim_{x\to 3}[x^2(x-1)]$

$$\lim_{x\to 3}[x^2(x-1)] = \lim_{x\to 3}(x^2) \cdot \lim_{x\to 3}(x-1)$$

$$\lim_{x\to 3}[x^2(x-1)] = (3^2) \cdot (3-1)$$

$$\lim_{x\to 3}[x^2(x-1)] = (9) \cdot (2)$$

$$\boxed{\lim_{x\to 3}[x^2(x-1)] = 18}$$

ans

QUOTIENT LAW

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\frac{\lim_{x\to a}f(x)}{\lim_{x\to a}g(x)}$$

If $\lim_{x\to a} g(x) \neq 0$

Evaluate $\lim_{x\to 1}\left(\frac{1}{x-1}\right)$

$$\lim_{x\to 1}\left(\frac{1}{x-1}\right)=\frac{\lim_{x\to 1}(1)}{\lim_{x\to 1}(x-1)}$$

$$\lim_{x\to 1}\left(\frac{1}{x-1}\right)=\frac{1}{1-1}$$

$$\lim_{x\to 1}\left(\frac{1}{x-1}\right)=\frac{1}{0}=\infty$$

$$\boxed{\lim_{x\to 1}\left(\frac{1}{x-1}\right)=\textit{does not exist}}$$

ans

QUOTIENT LAW

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\frac{\lim_{x\to a}f(x)}{\lim_{x\to a}g(x)}$$

If $\lim_{x\to a} g(x) \neq 0$

Evaluate $\lim_{x\to 2}\left(\frac{x-1}{x^2-1}\right)$

$$\lim_{x\to 2}\left(\frac{x-1}{x^2-1}\right)=\frac{\lim_{x\to 2}(x-1)}{\lim_{x\to 2}(x^2-1)}$$

$$\lim_{x\to 2}\left(\frac{x-1}{x^2-1}\right)=\frac{2-1}{2^2-1}$$

$$\lim_{x\to 2}\left(\frac{x-1}{x^2-1}\right)=\frac{1}{4-1}$$

$$\boxed{\lim_{x\to 2}\left(\frac{x-1}{x^2-1}\right)=\frac{1}{3}}$$

ans

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\frac{\lim_{x\to a}f(x)}{\lim_{x\to a}g(x)}$$

If $\lim_{x\to a} g(x) \neq 0$

Evaluate $\lim_{x\to 1}\left(\frac{x-1}{x^2-1}\right)$

$$\lim_{x\to 1}\left(\frac{x-1}{x^2-1}\right)=\lim_{x\to 1}\left[\frac{\cancel{(x-1)}^{\,1}}{(x-1)(x+1)}\right]$$

$$\lim_{x\to 1}\left(\frac{1}{x+1}\right)=\left(\frac{1}{1+1}\right)$$

$$\lim_{x\to 1}\left(\frac{1}{x+1}\right)=\frac{1}{2}$$

$$\boxed{\lim_{x\to 1}\left(\frac{x-1}{x^2-1}\right)=\frac{1}{2}}$$

ans

$$\lim_{x\to a}\left(f(x)\right)^n = \left(\lim_{x\to a} f(x)\right)^n$$

Evaluate $\lim_{x\to 2}\left(x^3 - 2x\right)^2$

$$\lim_{x\to 2}\left(x^3 - 2x\right)^2 = \left[\lim_{x\to 2}\left(x^3 - 2x\right)\right]^2$$

$$\lim_{x\to 2}\left(x^3 - 2x\right)^2 = \left[\left(2^3 - 2(2)\right)\right]^2$$

$$\lim_{x\to 2}\left(x^3 - 2x\right)^2 = [(8 - 4)]^2$$

$$\lim_{x\to 2}\left(x^3 - 2x\right)^2 = 4^2$$

$$\boxed{\lim_{x\to 2}\left(x^3 - 2x\right)^2 = 16}$$

ans

ROOT LAW

$$\lim_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to a} f(x)}$$

where n is a positive integer

Evaluate $\lim_{x \to 4} \sqrt{x+2}$

$$\lim_{x \to 4} \sqrt{x+2} = \sqrt{\lim_{x \to 4}(x+2)}$$

$$\lim_{x \to 4} \sqrt{x+2} = \sqrt{4+2}$$

$$\boxed{\lim_{x \to 4} \sqrt{x+2} = \sqrt{6}}$$

ans

SEATWORK